

Do Tech Layoffs Help Stock Prices?

Azam Ahmed and Xuan Liao · CS 210, Spring 2026

1. The Question

Every time a big tech company does layoffs, the news always says Wall Street loves it and the stock goes up. We wanted to actually check if that's true with data.

We grabbed every layoff event we could from layoffs.fyi (4,335 of them across 2,873 companies), pulled the matching stock prices from Yahoo Finance, and looked at what the stock actually did in the days right before and after each announcement. The short answer: the data does not back up the “Wall Street rewards layoffs” story. If anything, stocks dip a little. Mostly the market just doesn’t care that much.

2. The Data

Two sources, both free.

Source	Format	What we used
layoffs.fyi (Kaggle mirror)	CSV	4,335 layoff events
Yahoo Finance (yfinance)	JSON	281,418 daily price rows
S&P 500 (^GSPC)	JSON	1,666 trading days as benchmark

We used a frozen Kaggle copy of layoffs.fyi instead of scraping the live site so the results would be reproducible. Yahoo Finance handles stock splits and dividends through the adjusted close price, so we did not have to do anything special there.

3. The Database

Everything goes into Postgres. We have five tables in third normal form: **companies**, **layoff_events**, **daily_prices**, **market_index**, and **event_windows** (where the analysis results live). Prices use NUMERIC instead of FLOAT so chained log return calculations don’t accumulate rounding errors. CHECK constraints catch bad data at the database layer (no negative employee counts, no percentages over 100).

```
CREATE TABLE companies (  
  company_id      SERIAL PRIMARY KEY,  
  company_name    VARCHAR(255) NOT NULL,  
  ticker_symbol   VARCHAR(10),  
  industry        VARCHAR(100),  
  is_public       BOOLEAN NOT NULL DEFAULT FALSE,  
  CONSTRAINT uq_company_name UNIQUE (company_name),  
  CONSTRAINT uq_ticker      UNIQUE (ticker_symbol)  
);
```

Code 1. The companies table. Source: sql/schema.sql.

4. The Pipeline

Five Python scripts, one per stage. Each one is safe to re-run because everything uses ON CONFLICT, so if something fails halfway through we can just run it again.

- **Phase 1, ingest:** read the CSV, clean the columns, dedupe by (company, date), insert into companies and layoff_events.
- **Phase 2a, ticker matching:** figure out the stock ticker for each company.
- **Phase 2b, price fetch:** pull daily prices from Yahoo for every matched ticker.
- **Phase 3, event study:** compute returns, run the t-test, fit the regression and the Random Forest.
- **Phase 4, plots:** make the six figures.

Ticker matching turned out to be the hardest part. layoffs.fyi has “Meta”, “Meta Platforms”, and “Facebook” all listed as the same company. We tried fuzzy string matching first, but it kept giving us wrong answers (Apple matched to APLE, which is a hotel company called Apple Hospitality REIT). So we just made a manual map for the top 50 firms and used Yahoo’s search API for the rest.

```
MANUAL_TICKER_MAP = {
    "meta": "META", "facebook": "META",
    "google": "GOOGL", "alphabet": "GOOGL",
    "amazon": "AMZN", "microsoft": "MSFT",
    "apple": "AAPL", "netflix": "NFLX",
    "nvidia": "NVDA", "amd": "AMD",
    # ... ~50 more entries
}
```

Code 2. The manual ticker map. Source: src/match_tickers.py.

After matching we ended up with 196 public companies that we could actually pull prices for. That covers about 56% of all the laid-off employees in the dataset, even though it’s only ~7% of the companies. Most of the unmatched ones are private VC-funded startups.

5. The Analysis

For each layoff event we line it up with the trading calendar, then compute three things: the stock’s log return on each day, what the S&P 500 did that same day, and the difference between them (the “abnormal return”). Adding those up across the window around the announcement gives the Cumulative Abnormal Return, or CAR. We picked a 5-day window on either side as the main one, with a couple of others as backup.

```
market["r_mkt"] = np.log(market["adj_close"] / market["adj_close"].shift(1))

for cid, grp in prices.groupby("company_id"):
    g = grp.sort_values("trade_date").copy()
    g["r_stock"] = np.log(g["adj_close"] / g["adj_close"].shift(1))
    per_company[int(cid)] = g[["adj_close", "r_stock"]]
```

Code 3. Computing log returns. Source: src/analyze_events.py.

Then we run a one-sample t-test against zero, plus a 1,000-sample bootstrap to get a confidence interval, plus a standardized version of the test (called Patell-Z) which down-weights events where the stock was already volatile to begin with.

6. What We Found

Across 505 events with full data, the average CAR comes out to **negative 1.12%**. The plain t-test gives $p = 0.08$, which is borderline. The standardized version (Patell-Z) is more decisive at $p = 0.018$. Both are

negative, both lean against the “rewards” story.

	Value
Number of events	505
Mean CAR (-5, +5)	-1.12%
Median CAR	-0.05%
t-statistic	-1.75
p-value (t-test)	0.080
95% bootstrap CI	[-2.37%, +0.13%]
Patell-Z p-value	0.018

Table 1. Headline numbers for the (-5, +5) window.

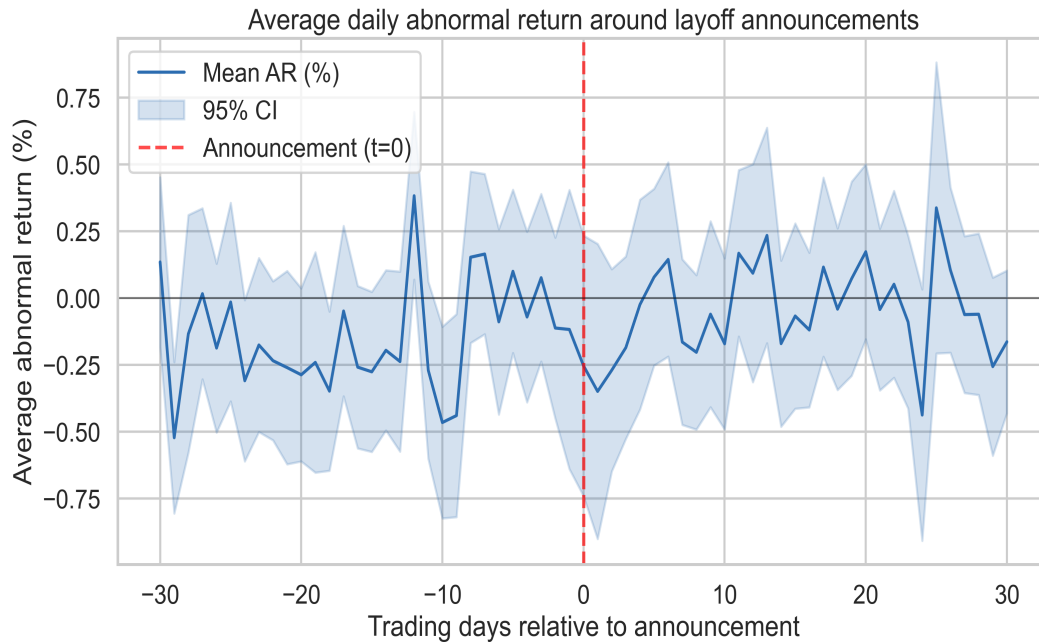


Figure 1. Average abnormal return per trading day, from 30 days before the announcement to 30 days after. The line sits a little below zero pretty much the whole time after the announcement, which is consistent with a small but persistent negative drift.

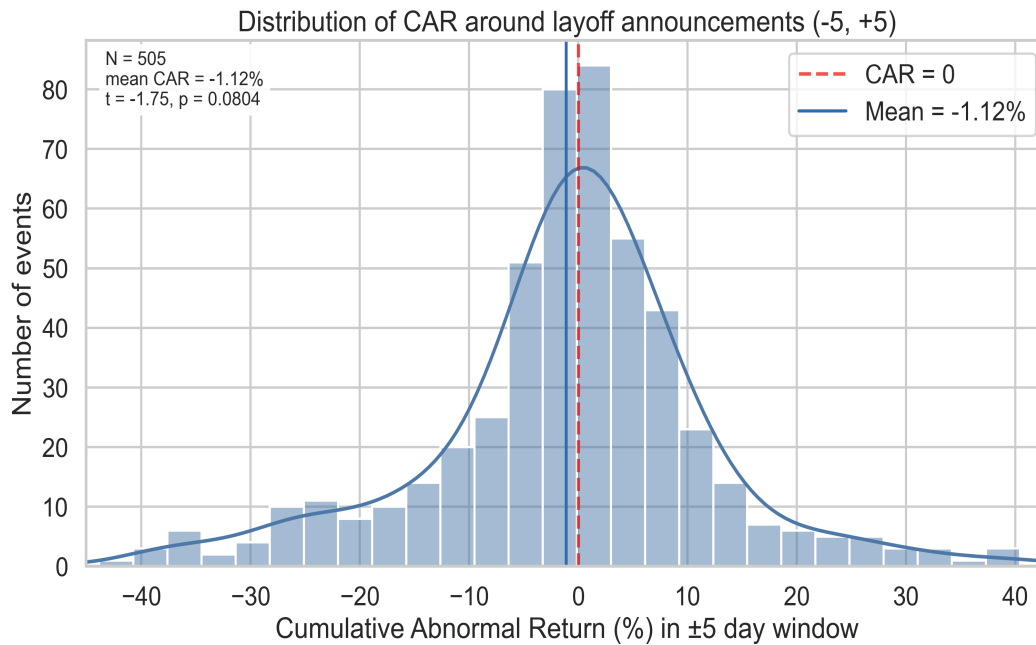


Figure 2. The distribution of CARs across all 505 events. It's a fat-tailed bell curve, but the bulk of it sits a little to the left of zero, which is what gives us the negative average.

7. Other Things We Looked At

Does layoff size matter?

Not really. We regressed CAR against log of employees laid off and got a slope of basically zero ($p = 0.99$). A 5,000-person cut gets the same average reaction as a 50-person cut. The percentage of workforce shows a tiny negative trend at the high end, but that's driven by a handful of outliers in small companies.

Layoff size vs. stock-price reaction

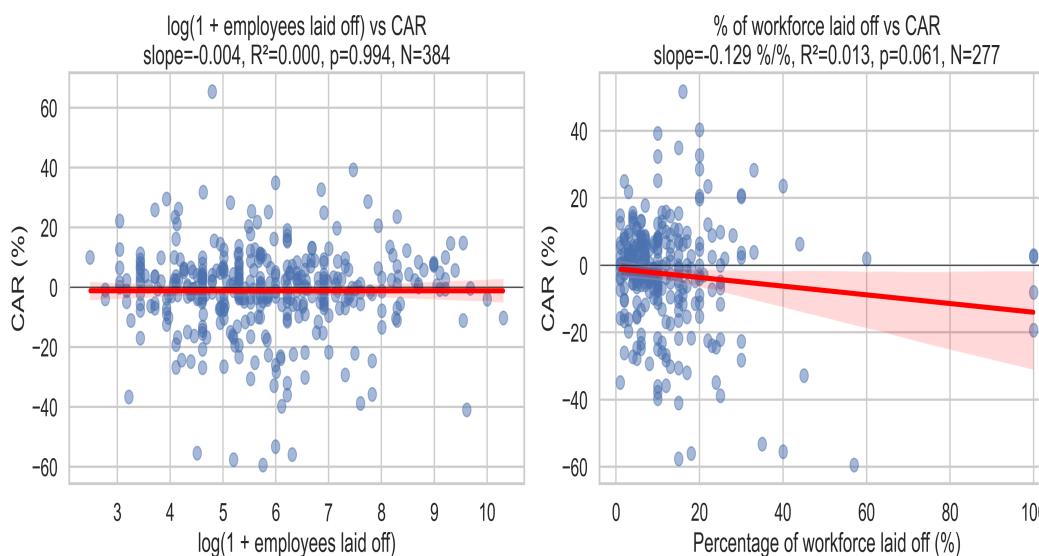


Figure 3. Layoff size vs CAR. Both lines are basically flat.

First-time vs repeat layoffs?

First-time layoffs are way more negative on average. Mean CAR is -2.99% for first-timers vs -0.16% for repeats (Welch t-test $p = 0.035$). Makes sense honestly. The first cut is real new information about the company. By the third or fourth round, the market already knows what's going on.

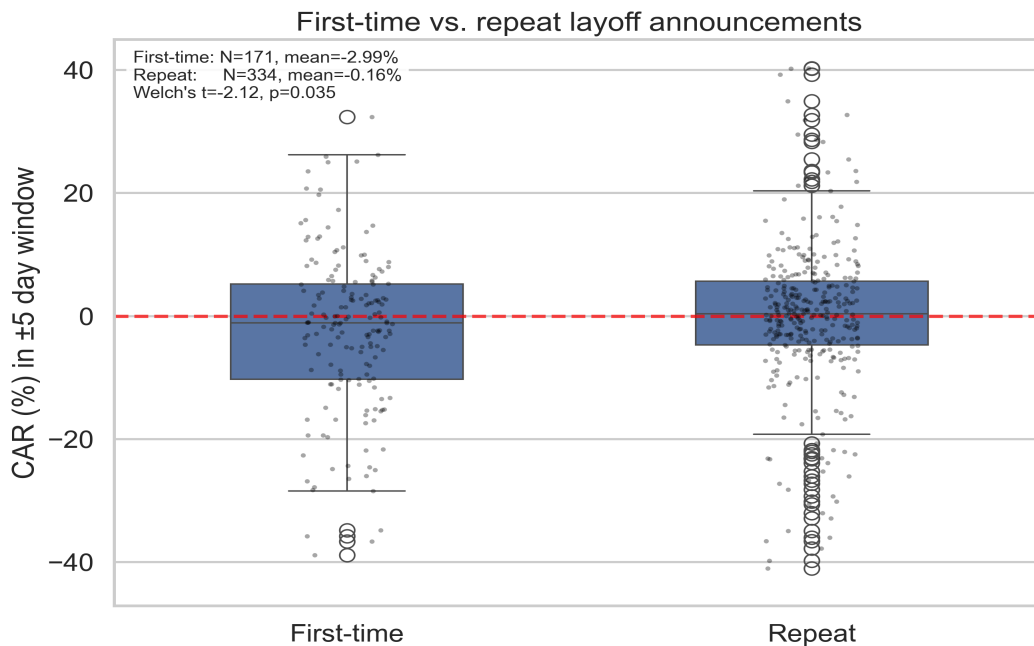


Figure 4. CAR for first-time vs repeat layoffs. Both distributions are wide, but the first-time mean is meaningfully lower.

Random Forest classifier.

We trained a Random Forest to predict whether a given layoff would have a positive or negative reaction. It got an AUC of 0.46 in 5-fold cross-validation, which is actually worse than random guessing. The most important feature was not even layoff size, it was the broader market regime (was the S&P up or down that month). Layoff-specific features just don't carry much signal.

```
clf = RandomForestClassifier(
    n_estimators=500,
    random_state=42,
    class_weight="balanced",
    n_jobs=-1,
)
auc_scores = cross_val_score(pipe, X, y, scoring="roc_auc", cv=5, n_jobs=-1)
```

Code 4. Random Forest setup. Source: `src/analyze_events.py`.

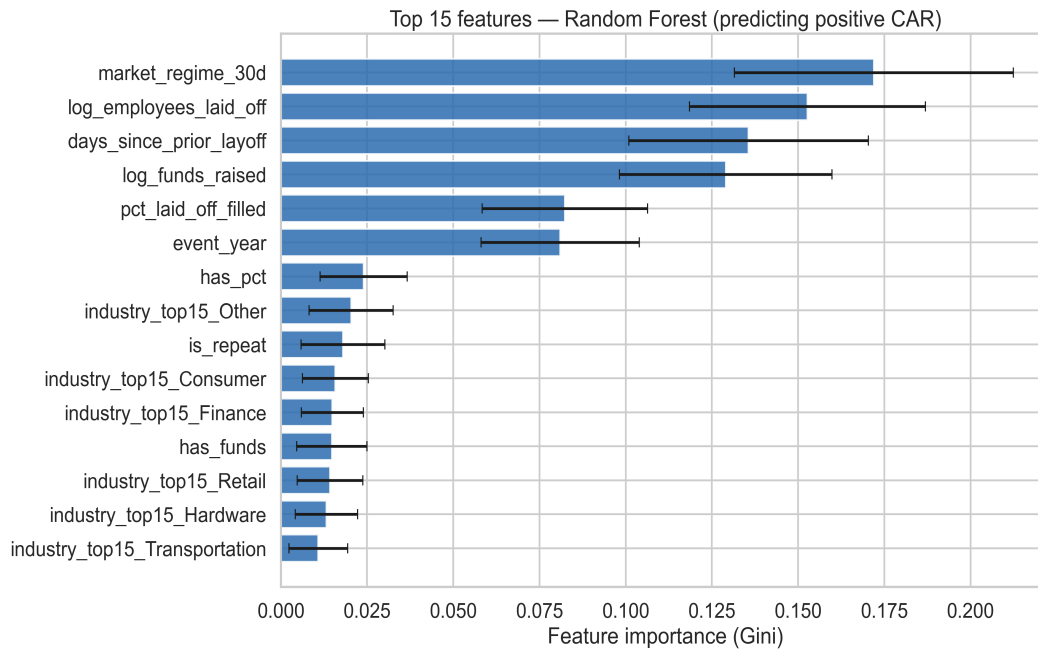


Figure 5. Feature importance from the Random Forest. Market regime is the top feature, everything else is roughly tied below it.

Layoffs over time.

January 2023 alone had 42 matched-company layoffs, the peak of the post-ZIRP correction. Mean CARs vary a lot month to month but with no clear pattern. The biggest outliers (like November 2021 at -38.8%) come from cells with only one or two events.

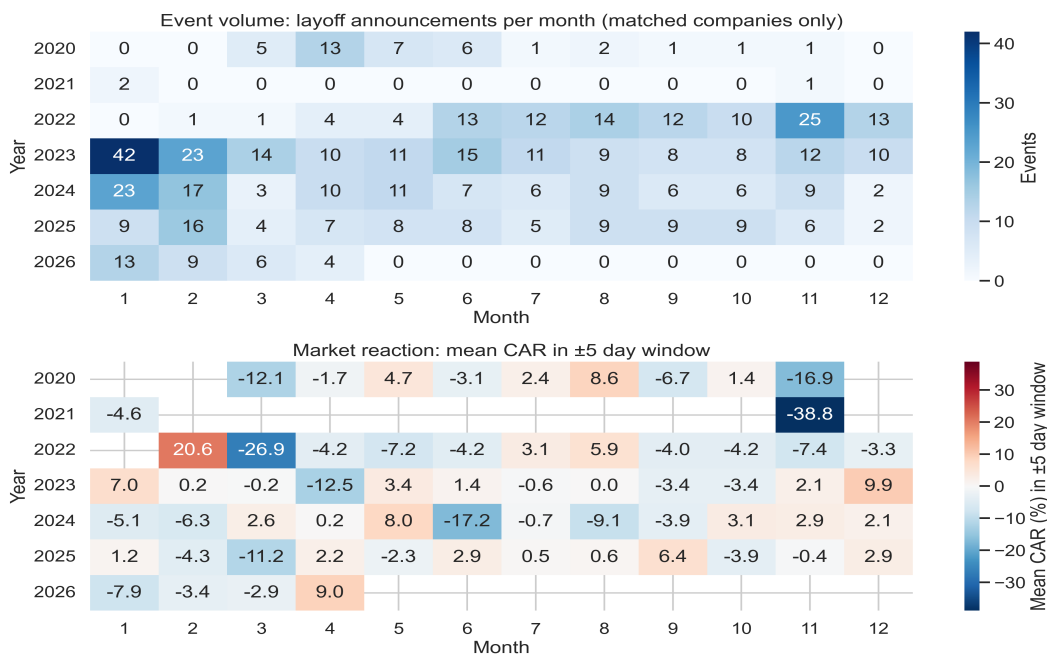


Figure 6. Top: how many layoffs each month. Bottom: average CAR each month.

8. So What Does This Mean

Two stories that get told about layoffs:

- **“Wall Street rewards layoffs.”** This would predict a positive mean CAR with a small p-value. We got the opposite. So this story is not really supported.
- **“Wall Street punishes layoffs.”** This would predict a big negative effect. We see a small negative effect (~1%), so this is only weakly supported.

The way we’d actually describe it: **by the time a layoff hits the news, the market already knows.** The underlying weakness in the business has usually shown up in earnings calls or sector trends already. The layoff is more of a confirmation than actual news, which is why we see a small negative tilt instead of a big one.

9. Limitations

- **Sample bias.** Only ~7% of companies in the dataset are public with tickers. The rest are private startups we had to skip. Our results don’t generalize to those.
- **One benchmark.** We compare against the S&P 500 only. A multi-factor model would be better but probably wouldn’t change the direction.
- **Leak dates.** Some layoffs leak to the press before the official announcement. If that happens, our window misses the early reaction and underestimates the size of the effect.
- **Non-US companies.** A small fraction of the matched companies aren’t US listed, which adds a little noise from currency and exchange calendar mismatches.

10. Who Did What

Azam Ahmed (Phases 2b, 3, 4)

- Built the price fetcher (yfinance with retry logic).
- Wrote the event-study code: log returns, CAR, SCAR.
- Ran the t-test, bootstrap, and Patell-Z test.
- Built the OLS regression and the Random Forest.
- Made all six figures and the result tables.

Xuan Liao (Phases 1, 2a, documentation)

- Designed the Postgres schema with constraints and indexes.
- Wrote the ingest script (CSV cleaning, dedup, two-pass insert).
- Built the manual ticker override map for the top 50 firms.
- Wrote the Phase 2a ticker matcher (Yahoo search + name validation).
- Wrote the project docs: README, Phase 2a write-up, SQL migrations.

Both of us reviewed all the code and worked on the report and demo together.

References

- MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13-39.

- Worrell, D. L., Davidson, W. N., & Sharma, V. M. (1991). Layoff announcements and stockholder wealth. *Academy of Management Journal*, 34(3), 662-678.
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- layoffs.fyi tracker, Kaggle mirror: kaggle.com/datasets/swaptr/layoffs-2022.
- Yahoo Finance via the *yfinance* Python package.